

HARTNES STATE, VT, USA

WMO: 726115

Lat: 43.344N

Lon: 72.518W

Elev: 578

StdP: 14.39

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 54740

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.7	(c) 0.5	(d) -15.3	(e) 2.4	(f) -1.1	(g) -10.6	(h) 3.2	(i) 2.7	(j) 20.2	(k) 24.8	(l) 18.3	(m) 25.3	(n) 2.4	(o) 300	(p) 0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 22.3	(c) 88.3	(d) 70.9	(e) 85.2	(f) 69.0	(g) 82.3	(h) 67.8	(i) 74.3	(j) 82.8	(k) 72.4	(l) 80.4	(m) 70.8	(n) 78.3	(o) 8.0	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.0	121.0	77.8	70.2	113.4	76.0	68.3	106.3	74.6	38.1	82.7	36.4	80.2	35.0	78.5	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	15.0	12.4	-13.7	93.9	5.5	4.6	-17.7	97.3	-20.9	100.0	-24.0	102.6	-27.9	106.0
			-14.0	77.8	5.3	2.8	-17.8	79.8	-21.0	81.5	-24.0	83.1	-27.8	85.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.1	20.9	23.7	31.5	44.4	56.1	64.5	69.7	68.0	60.7	48.2	37.3	27.3		
	DBStd	18.86	11.00	9.87	9.42	8.21	7.76	6.46	5.25	5.24	7.16	7.80	8.36	9.13		
	HDD50	3683	904	736	577	209	27	0	0	0	5	130	390	705		
	HDD65	7298	1369	1156	1038	620	294	87	16	29	167	521	832	1169		
	CDD50	2273	0	0	4	40	215	434	610	559	326	76	8	1		
	CDD65	412	0	0	0	1	17	71	161	123	38	1	0	0		
	CDH74	4104	0	0	9	58	300	749	1547	1075	346	20	0	0		
	CDH80	1189	0	0	1	15	79	216	507	281	89	1	0	0		
(14) Wind	WSAvg	3.8	4.1	4.4	4.7	4.9	4.2	3.5	3.2	2.8	2.8	3.5	3.8	3.7		
(15) Precipitation	PrecAvg	44.1	3.3	2.8	3.6	3.8	4.0	4.1	3.7	3.9	3.8	3.8	4.1	3.6		
	PrecMax	56.3	8.0	9.1	8.3	8.2	11.3	12.2	7.2	8.9	8.4	12.4	7.2	8.1		
	PrecMin	30.3	0.4	0.2	0.7	0.7	1.0	0.7	1.0	0.8	0.3	0.4	1.1	1.4		
	PrecStd	6.4	1.9	1.4	1.5	1.6	2.0	2.4	1.7	1.8	1.9	2.4	1.6	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.4	53.5	70.2	81.2	86.5	90.6	92.8	90.2	88.0	76.9	65.0	55.0		
		MCWB	44.2	43.6	55.1	60.7	65.7	72.6	73.7	72.1	71.0	63.4	58.1	49.4		
	2%	DB	42.2	45.4	56.6	72.4	81.8	86.2	88.6	85.9	82.0	70.5	60.7	47.0		
		MCWB	37.7	38.1	46.4	55.3	63.4	69.6	71.4	69.9	68.2	60.3	54.1	42.4		
	5%	DB	37.6	40.8	49.8	66.1	76.7	81.9	85.6	83.0	77.6	65.6	54.8	43.1		
		MCWB	33.8	35.3	41.4	52.5	61.0	67.8	69.8	69.1	66.0	58.2	49.1	38.7		
	10%	DB	35.2	36.8	45.0	60.9	72.1	78.7	82.2	80.6	73.4	61.3	50.4	38.8		
		MCWB	32.3	32.5	38.8	48.8	58.6	64.9	68.2	67.5	64.0	54.8	45.1	35.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.5	44.4	57.0	63.7	69.5	76.9	76.6	75.8	74.1	67.2	61.1	50.7		
		MCDB	49.0	49.6	68.5	73.5	80.6	85.8	86.8	84.3	81.2	71.3	64.4	54.9		
	2%	WB	38.3	39.3	47.8	58.9	66.7	73.2	74.2	73.3	71.1	62.5	55.3	44.0		
		MCDB	41.5	44.1	53.8	69.7	76.9	81.8	83.4	80.7	77.7	67.8	59.2	46.7		
	5%	WB	34.9	35.8	42.7	54.3	64.0	70.4	72.5	71.5	68.8	59.5	50.4	39.4		
		MCDB	37.0	39.8	48.5	62.8	72.2	77.8	80.9	79.1	74.9	64.9	54.6	42.2		
	10%	WB	32.8	33.5	39.1	50.1	61.3	67.9	70.8	69.9	66.4	56.1	46.0	35.9		
		MCDB	34.7	35.9	44.5	59.7	69.1	74.8	78.7	76.3	72.1	60.3	49.1	38.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.9	18.5	18.7	22.0	23.0	21.8	22.3	22.2	22.3	20.0	16.7	14.6		
		MCDBR	17.7	21.9	27.4	34.1	31.7	27.1	26.3	25.2	26.7	26.3	24.0	18.6		
	5% WB	MCWBR	14.6	16.1	17.6	18.5	15.8	13.2	11.2	11.3	14.2	16.8	17.9	15.1		
		MCWBR	16.6	19.7	24.9	29.4	25.5	23.0	22.3	21.0	21.7	22.6	21.4	17.3		
(40) Clear-Sky Solar Irradiance	taub		0.289	0.294	0.321	0.355	0.387	0.427	0.446	0.420	0.370	0.338	0.308	0.284		
		taud	2.384	2.348	2.396	2.397	2.348	2.253	2.207	2.265	2.393	2.492	2.525	2.478		
	Ebn at Noon		268	287	292	289	281	269	262	265	272	268	261	261		
		Edh at Noon	25	30	33	36	39	43	44	40	33	26	22	21		
(44) All-Sky Solar Radiation	RadAvg		510	790	1158	1428	1596	1718	1788	1616	1311	813	548	394		
	RadStd		45	72	77	168	186	158	124	87	93	86	59	41		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (58 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+155	+59

Nomenclature: See separate page